

Q1 - 25 July - Shift 1

If $\lim_{n \rightarrow \infty} (\sqrt{n^2 - n - 1} + n\alpha + \beta) = 0$ then $8(\alpha + \beta)$

is equal to :

(A) 4

(B) -8

(C) -4

(D) 8

Space for your notes:

Q2 - 25 July - Shift 1

If $\lim_{n \rightarrow \infty} \frac{(n+1)^{k-1}}{n^{k+1}} [(nk + 1) + (nk + 2) + \dots +$

$(nk + n)] = 33$. $\lim_{n \rightarrow \infty} \frac{1}{n^{k+1}} \cdot [1^k + 2^k + 3^k + \dots + n^k]$,

then the integral value of k is equal to _____.

Space for your notes:

Q3 - 25 July - Shift 2

$\lim_{x \rightarrow \frac{\pi}{4}} \frac{8\sqrt{2} - (\cos x + \sin x)^7}{\sqrt{2} - \sqrt{2} \sin 2x}$ is equal to

(A) 14

(B) 7

(C) $14\sqrt{2}$

(D) $7\sqrt{2}$

Space for your notes:

Q4 - 26 July - Shift 1

If the function

$$f(x) = \begin{cases} \frac{\log_e(1-x+x^2) + \log_e(1+x+x^2)}{\sec x - \cos x}, & x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) - \{0\} \\ k, & x=0 \end{cases}$$

is continuous at $x = 0$, then k is equal to :

- (A) 1 (B) -1
(C) e (D) 0

Space for your notes:

Q5 - 26 July - Shift 2

Let $\beta = \lim_{x \rightarrow 0} \frac{\alpha x - (e^{3x} - 1)}{\alpha x (e^{3x} - 1)}$ for some $\alpha \in \mathbb{R}$. Then

the value of $\alpha + \beta$ is :

- (A) $\frac{14}{5}$ (B) $\frac{3}{2}$ (C) $\frac{5}{2}$ (D) $\frac{7}{2}$

Space for your notes:

Q6 - 28 July - Shift 1

$$\lim_{x \rightarrow 0} \left(\frac{(x + 2 \cos x)^3 + 2(x + 2 \cos x)^2 + 3 \sin(x + 2 \cos x)}{(x + 2)^3 + 2(x + 2)^2 + 3 \sin(x + 2)} \right)^{\frac{100}{x}}$$

is equal to _____.

Space for your notes:

Q7 - 29 July - Shift 1

If $\lim_{x \rightarrow 0} \frac{\alpha e^x + \beta e^{-x} + \gamma \sin x}{x \sin^2 x} = \frac{2}{3},$

Space for your notes:

where $\alpha, \beta, \gamma \in \mathbb{R}$, then which of the following is

NOT correct ?

(A) $\alpha^2 + \beta^2 + \gamma^2 = 6$

(B) $\alpha\beta + \beta\gamma + \gamma\alpha + 1 = 0$

(C) $\alpha\beta^2 + \beta\gamma^2 + \gamma\alpha^2 + 3 = 0$

(D) $\alpha^2 - \beta^2 + \gamma^2 = 4$

Answer Key

Q1 (C)

Q2 (5)

Q3 (A)

Q4 (A)

Q5 (C)

Q6 (1)

Q7 (C)

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Q1 (C)

$$\lim_{n \rightarrow \infty} n \left(1 - \frac{n+1}{n^2} \right)^{\frac{1}{2}} + \alpha n + \beta = 0$$

$$\lim_{n \rightarrow \infty} n \left\{ 1 - \frac{1}{2} \left(\frac{n+1}{n^2} \right) + \frac{\left(\frac{1}{2} \right) \left(-\frac{1}{2} \right)}{2!} \left(\frac{n+1}{n^2} \right)^2 + \dots \right\} + \alpha n + \beta = 0$$

$$\lim_{n \rightarrow \infty} n - \frac{1}{2} + \frac{1}{2n} + \dots + n\alpha + \beta = 0$$

$$\alpha = -1, \beta = \frac{1}{2}$$

$$8(\alpha + \beta) = -4$$

Q2 (5)

LHS

$$\lim_{n \rightarrow \infty} \frac{(n+1)^{k-1}}{n^{k+1}} [nk \cdot n + 1 + 2 + \dots + n]$$

$$= \lim_{n \rightarrow \infty} \frac{(n+1)^{k-1}}{n^{k+1}} \cdot \left[n^2 k + \frac{n(n+1)}{2} \right]$$

$$= \lim_{n \rightarrow \infty} \frac{(n+1)^{k-1} \cdot n^2 \left(k + \frac{\left(1 + \frac{1}{n}\right)}{2} \right)}{n^{k+1}}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right) \left(k + \frac{\left(1 + \frac{1}{n}\right)}{2} \right)$$

$$\Rightarrow \left(k + \frac{1}{2} \right)$$

RHS

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{1}{n^{k+1}} (1^k + 2^k + \dots + n^k) = \frac{1}{k+1}$$

LHS = RHS

$$\Rightarrow k + \frac{1}{2} = 33 \cdot \frac{1}{k+1}$$

$$\Rightarrow (2k+1)(k+1) = 66$$

$$\Rightarrow (k-5)(2k+13) = 0$$

Q3 (A)

(A)

Q4 (A)

$$\lim_{x \rightarrow 0} \frac{(\ln(1+x^2+x^4)) \cos x}{1 - \cos^2 x}$$

$$\lim_{x \rightarrow 0} \frac{\left(\frac{\ln(1+x^2+x^4)}{x^2+x^4} \right) x^2 (1+x^2) \cos x}{\left(\frac{\sin^2 x}{x^2} \right) x^2} = 1$$

$$\therefore k = 1$$

Q5 (C)

$$\beta = \lim_{x \rightarrow 0} \frac{\alpha x - (e^{3x} - 1)}{\alpha x (e^{3x} - 1)}$$

$$\beta = \lim_{x \rightarrow 0} \frac{1 + \alpha x - \left[1 + 3x + \frac{9x^2}{2!} + \dots \right]}{(\alpha x) \frac{(e^{3x} - 1)}{3x}}$$

$$\beta = \lim_{x \rightarrow 0} \frac{(\alpha x - 3x) - \frac{9x^2}{2!} - \dots}{3\alpha x^2}$$

For existence of limit $\alpha - 3 = 0$

$$\alpha = 3$$

$$\text{Limit } \beta = \frac{-3}{2\alpha}$$

$$\beta = -\frac{1}{2}$$

Now,

$$\alpha + \beta = \frac{5}{2}$$

Q6 (1)

$$\lim_{x \rightarrow 10} \left(\frac{(x+2\cos x)^3 + 2(x+2\cos x)^2 + 3\sin(x+2\cos x)}{(x+2)^3 + 2(x+2)^2 + 3\sin(x+2)} \right)^x$$

Form 1^∞

$$= e^{\lim_{x \rightarrow 0} \left[\left(\frac{(x+2\cos x)^3 + 2(x+2\cos x)^2 + 3\sin(x+2\cos x)}{(x+2)^3 + 2(x+2)^2 + 3\sin(x+2)} \right) - 1 \right] \times \frac{100}{x}}$$

$$= e^{\lim_{x \rightarrow 0} \left[\frac{100}{x} \left(\frac{(x+2\cos x)^3 + 2(x+2\cos x)^2 + 3\sin(x+2\cos x) - ((x+2)^3 + 2(x+2)^2 + 3\sin(x+2))}{(x+2)^3 + 2(x+2)^2 + 3\sin(x+2)} \right) \right]}$$

$$= e^{\lim_{x \rightarrow 0} \frac{100}{x} \left[\frac{(x+2\cos x)^3 + (x+2)^3 + 2(x+2\cos x)^2 - 2(x+2)^2 + 3\sin(x+2\cos x) - 3\sin(x+2)}{8+8+3\sin^2} \right]}$$

$$= e^{\frac{100}{16+3\sin^2} \lim_{x \rightarrow 0} \frac{3(x+2\cos x)^2 \times (1+2\sin x) - 3(x+2)^2 - 4(x+2\cos x)}{x(1-2\sin x) - 4(x+2) + 3\cos(x+2\cos x) \times (1-2\sin x) - 3\cos(x+2)}}$$

$$= e^{\frac{100}{16+3\sin^2} \left(\frac{12 - 3(4) + 8 \times 1 - 8 + 3\cos 2 - 3\cos 2}{1} \right)}$$

Using L'H rule.

$$= e^0 = 1$$

Q7 (C)

$$\lim_{x \rightarrow 0} \frac{\alpha \left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots\right) + \beta \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots\right) + \gamma \left(x - \frac{x^3}{3!} + \dots\right)}{x^3}$$

constant terms should be zero

$$\Rightarrow \alpha + \beta = 0$$

coeff of x should be zero

$$\Rightarrow \alpha - \beta + \gamma = 0$$

coeff of x^2 should be zero

$$\lim_{x \rightarrow 0} \frac{x^3 \left(\frac{\alpha}{3!} - \frac{\beta}{3!} - \frac{\gamma}{3!} \right) + x^4 \left(\frac{\alpha}{3!} - \frac{\beta}{3!} - \frac{\gamma}{3!} \right)}{x^3} = \frac{2}{3}$$

$$\Rightarrow \frac{\alpha}{2} + \frac{\beta}{2} = 0$$

$$\frac{\alpha}{6} - \frac{\beta}{6} - \frac{\gamma}{6} = 2/3$$

$$\Rightarrow \alpha = 1, \beta = -1, \gamma = -2$$